Local Failure After Episcleral Brachytherapy for Posterior Uveal Melanoma: Patterns, Risk Factors, and Management

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• PURPOSE: To evaluate the patterns, the risk factors, and the management of recurrence following brachytherapy in patients with posterior uveal melanoma, given that an understanding of the recurrence patterns can improve early recognition and management of local treatment failure in such patients.

• DESIGN: Retrospective cohort study.

• METHODS: SETTING: Multispecialty tertiary care center. PARTICIPANTS: A total of 375 eyes treated with episcleral brachytherapy for posterior uveal melanoma from January 2004 to December 2014. Exclusion criteria included inadequate follow-up (<1 year) and previous radiation therapy. MAIN OUTCOMES AND MEASURES: Local control rate and time to recurrence were the primary endpoints. Kaplan-Meier estimation and Cox proportional hazards models were conducted to identify risk factors for recurrence.

• RESULTS: Twenty-one patients (5.6%) experienced recurrence (follow-up range 12–156 months; median 47 months). The median time to recurrence was 18 months (range 4–156 months). Five-year estimated local recurrence rate was 6.6%. The majority (90.5%) of the recurrences occurred within the first 5 years. The predominant site of recurrence was at the tumor margin (12 patients, 57.1%). Univariate analysis identified 3 statistically significant recurrence risk factors: advanced age, largest basal diameter, and the use of adjuvant transpupillary thermotherapy (TTT). Recurrent tumors were managed by repeat brachytherapy, TTT, or enucleation.

• CONCLUSIONS: Local recurrences following brachytherapy are uncommon 5 years after episcleral brachytherapy. Follow-up intervals can be adjusted to reflect time to recurrence. Most of the eyes with recurrent tumor can be salvaged by conservative methods. (Am J Ophthalmol 2017;177:9–16. © 2017 Elsevier Inc. All rights reserved.)

THE MULTICENTER RANDOMIZED CLINICAL TRIAL conducted by the Collaborative Ocular Melanoma Study (COMS) showed equivalent 5-year mortality rates in patients with medium-sized choroidal melanoma treated by iodine-125 (^{125}I) episcleral brachytherapy or by enucleation.[1] Since then, episcleral brachytherapy has become the most frequent modality of treatment for posterior uveal melanoma.[2,3] Despite high initial control rates, an average of 9.6% of patients with uveal melanoma experience local recurrence following brachytherapy.[4–9]

To date, few studies have published information about management options of local recurrence after failed brachytherapy to preserve globe and vision. Damato and associates analyzed 458 patients treated by ruthenium-106 (^{106}Ru) brachytherapy and showed that eye preservation was achieved in 7 of the 9 cases of recurrent uveal melanoma.[10] Additional conservative treatments included repeat brachytherapy, proton beam radiation therapy, and transpupillary thermotherapy (TTT). In a retrospective analysis of 7 patients with uveal melanoma initially treated with ^{125}I brachytherapy, a repeat ^{125}I or ^{106}Ru brachytherapy had failed to control local tumor recurrence in only 1 case.[11] Similarly, Grange and associates reported a series of 13 patients treated with a repeat ^{106}Ru brachytherapy for nonresponse or recurrence of uveal melanoma and secondary treatment was successful in 11 of them over a follow-up up to 5 years, suggesting that brachytherapy is an effective therapeutic alternative to enucleation.[12]

Nevertheless, there are no clear data regarding the time of presentation and the location of tumor recurrences after brachytherapy in patients with posterior uveal melanoma. An understanding of the recurrence patterns may be beneficial for determining the appropriate treatment and the optimal length and frequency of the surveillance strategies. Early recognition of local tumor recurrence is also crucial, considering the increased risk of metastatic disease associated with local treatment failure.[1,13,14]

The aims of this study were to determine local failure rate, patterns (chronological and anatomic), and risk factors of recurrence in patients treated by episcleral brachytherapy for posterior uveal melanoma, with or without

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adjuvant TTT in a single referral center. The management options of recurrent cases and their outcomes were also analyzed.

METHODS

WE OBTAINED APPROVAL FROM THE INSTITUTIONAL Review Board of the Cleveland Clinic Foundation for this retrospective cohort study. Following standard-of-care guidelines, patients with large tumors, poor visual acuity, poor visual prognosis, optic nerve extension, and preference for enucleation rather than radiation were treated by enucleation.

We reviewed the clinical records of all the patients who underwent brachytherapy with ^{106}Ru or ^{125}I plaque for posterior uveal melanoma (ciliary body or choroidal) at the Department of Ophthalmic Oncology, Cole Eye Institute of the Cleveland Clinic between January 1, 2004 and December 31, 2014. The patients who received adjuvant TTT (in addition to brachytherapy) were included. The exclusion criteria were patients previously treated by radiation or who had less than 1 year of follow-up. Tumor and treatment parameters analyzed included largest basal diameter (LBD), apical height, distance from the posterior tumor margin to fovea and optic disc, location of recurrence, type of isotope, radiation dose to the apex, and the use of adjuvant TTT.

The initial diagnosis and tumor size (basal dimensions and height) of posterior uveal melanoma were determined by ophthalmic examination, ultrasonography (A- and B-scan), and fundus photography. Tumors were treated by ^{125}I plaque (model MED3631-A/M; North American Scientific, Chatsworth, California, USA, or Isoaid Advantage model IAI-125A; Isoaid, Port Richey, Florida, USA) or ^{106}Ru plaque (models CCA, CCD, and CCB, with diameters of 15.3, 17.9, and 20.2 mm, respectively; Bebig GmbH, Berlin, Germany) with at least a 2-mm safety margin in all directions. The radiation dose was calculated with a plaque simulator treatment planning software (Plaque Simulator V5.3.4; Bebig GmbH) using the tumor basal dimensions and apical height. Before the introduction of intraoperative ultrasonographic confirmation (ophthalmic ultrasound unit Ellex Eye Cubed V2; Minneapolis, Minnesota, USA, or Aviso; Quantel, Bozeman, Montana, USA) in June 2006, plaque positioning was guided by indirect ophthalmoscopy or transillumination. TTT was applied at the time of plaque removal and repeated at 3-month intervals postoperatively (total of 3 sessions), with the goal of achieving a gray coloration on the tumor surface after a 1-minute exposure.

All patients were followed at 1–2 weeks after brachytherapy, every 3 months for the first year, and then at 6-month intervals for a total of 5 years. Thereafter, surveillance is usually performed annually. Recorded outcome measures included local tumor recurrence rate, time to recurrence, and time to metastasis (for recurrent cases). Local recurrence was determined by a tumor growth after observed regression post-brachytherapy (increase of >0.5 mm in base or >0.2 mm in height from previous examination). All patients were offered systemic staging preoperatively (computed tomography chest, abdomen, and pelvis), followed by surveillance for metastases that included liver ultrasound every 6 months for the first 5 years after diagnosis and then annually.

• **STATISTICAL ANALYSIS:** Categorical factors were described using frequencies and percentages, while continuous measures were summarized as median, mean, and standard deviation. Comparisons of groups with and without recurrence were made using Pearson χ^2 tests or 2-sample t tests, as appropriate. Kaplan-Meier estimation and Cox proportional hazards models were used to identify potential predictors of local recurrence. Analyses were performed using SAS software (version 9.3; SAS, Cary, North Carolina, USA). Owing to the low number of recurrences, multivariate analysis was not performed.

• **LITERATURE REVIEW:** Review of published literature in English revealed that numerous studies have examined local tumor control in patients with uveal melanoma treated by brachytherapy. Because of a previous review of the data performed by Finger[2] and COMS,[1] we only included studies that reported predictors and options of management for local treatment failure following brachytherapy from 1998 onwards. The data including the number of patients, type of isotope, recurrence rate at 5 years and, risk factors and treatments for recurrence in some of these published studies is summarized (Table 1).

RESULTS

OF 416 PATIENTS TREATED BY BRACHYTHERAPY WITH OR without adjuvant TTT for primary posterior uveal melanoma, 42 patients were excluded because follow-up was less than 1 year. Overall, 374 patients (375 eyes) met inclusion criteria and were analyzed in this study. One patient had bilateral posterior uveal melanoma. The patient demographics, tumor characteristics, and treatment parameters are summarized in Table 2. The median age of diagnosis was 62 years (mean 61.5 years; range 16–92 years). The median LBD was 11 mm (mean 11 mm; range 2.5–20 mm) and the median tumor apical height was 3.1 mm (mean 3.9 mm; range 1–13 mm). Based on COMS criteria, 99 (26.4%) of the tumors were classified as small sized, 252 (67.2%) medium sized, and 24 (6.4%) large sized. Intraoperative plaque placement confirmation by ophthalmic ultrasound was performed in 316 cases (84.3%) The median delivered radiation dose to tumor apex for both ^{106}Ru and ^{125}I plaques was 85.2 Gy (mean 85.3 Gy; range 72.4–100.0 Gy) and 85.0 Gy (mean 84.7 Gy; range

TABLE 1. Characteristics of 9 Studies Reporting Risk Factors and Treatments of Recurrence Following Brachytherapy (1998–2016)

Authors	Year	N	Isotope	Recurrence Rate at 5 Years	Risk Factors	Treatment of Recurrence
Gunduz et al[15]	1999	630[a]	125-I/106-Ru/ 60-Cobalt/ 192-Iridium	9%	<2 mm from optic disc, presence of retinal invasion	Enucleation, brachytherapy
Lommatzsch et al[16]	2000	140	106-Ru/ 106-Rh	Not mentioned	Greater tumor dimensions	Not mentioned
Jampol et al[1]	2002	650	125-I	10.3%	Greater tumor dimensions, overlying retinal detachment, lower radiation dose to apex, advanced age, proximity to FAZ	Enucleation, additional local treatment
Rouberol et al[5]	2004	213	106-Ru	21.7%	Greater tumor dimensions, Bruch membrane rupture	Enucleation, brachytherapy
Damato et al[10]	2005	458	106-Ru	2.1%	Greater tumor dimensions	Enucleation, brachytherapy EBRT, TTT, combination
Bergman et al[17]	2005	579	106-Ru	16.6%[c]	Greater tumor dimensions	Enucleation, brachytherapy
Finger[2]	2009	400	103-Pd	3.3%	None	Enucleation
Papageorgious et al[18]	2011	189	106-Ru	Not mentioned	Greater tumor dimension, presence of retinal detachment	Not mentioned
Barker et al[6]	2014	28	106-Ru	41.5%	Low baseline acuity, proximity to optic disc and FAZ, posterior tumors, small-sized plaque, plaque-tumor <6 mm	Enucleation, TTT, PDT, proton beam therapy
Perez et al[8]	2014	190	125-I	9%	None	Not mentioned
Vonk et al[7]	2015	116	125-I	4.7%	None	Not mentioned
OOTF[13]	2016	3217[b]	NA	7.3%	None	Not mentioned

EBRT = external beam radiation therapy; FAZ = foveal avascular zone; OOTF = Ocular Oncology Task Force; PDT = photodynamic therapy; TTT = transpupillary thermotherapy.

[a]Choroidal melanoma with macular involvement.

[b]Includes other methods of treatment (TTT, resection).

[c]Enucleation rate including nonrecurrence cases.

72.5–96.7 Gy), respectively. During the follow-up time, 13 patients (3.5%) underwent enucleation due to neovascular glaucoma induced by radiation. The follow-up period ranged from 12 to 156 months, with a median of 47 months.

• **LOCAL RECURRENCE:** Local tumor recurrence was observed in 21 patients (5.6%). Most of the local recurrence occurred in women (66.7%). The median age at the time of recurrence was 69 years (mean 67.9 years; range 49–82 years). Seventeen patients (81%) were initially treated with ^{125}I plaque and 4 patients (19%) with ^{106}Ru plaque. The recurrent tumors had a median initial LBD of 13 mm (mean 12.9 mm; range 5–20 mm) and a median apical height of 4 mm (mean 4.4 mm; range 1.5–9 mm). Among the recurrent cases, the percentages of patients who had small-, medium-, and large-sized melanoma were 19.0%, 66.7%, and 14.3%, respectively (Table 2). The median time to recurrence was 18 months (range 4–156 months). The estimated 5-year local recurrence rate was 6.6% (Figure 1).

• **PATTERNS OF RECURRENCE:** The chronological distribution of local tumor failure could be divided into 4 categories: nonresponse, and early, delayed, and late recurrence. Among the 21 patients who failed brachytherapy, 2 patients (9.5%) experienced nonresponse within 3 months. Early recurrence within 3–12 months after treatment was observed in 5 patients (23.8%). Delayed recurrence (1–5 years postoperatively) was noted in 12 patients (57.1%), whereas late recurrence beyond 5 years occurred in 2 patients (9.5%). Therefore, 33.3%, 71.4%, and 90.5% of recurrence occurred within 1, 2, and 5 years, respectively (Figure 1).

The anatomic location of recurrences was classified as marginal (anterior [Figure 2], posterior, or temporal), central, diffuse, ring, or extraocular (Figure 3). The most common location of recurrence was at the tumor margins (12 patients; 57.1%): 6 patients with posterior marginal recurrence, 5 patients with anterior marginal recurrence, and 1 patient with temporal marginal recurrence. Tumor recurrence was central in 3 patients and diffuse in 3 patients. Two patients had developed ring pattern recurrence in the ciliary body and extraocular extension was observed in 1 patient.

• **RISK FACTORS:** Among univariate continuous predictors of recurrence, older age and greater LBD were identified as

TABLE 2. Patient Demographics, Tumor Characteristics, and Treatment Parameters of 375 Eyes With Posterior Uveal Melanoma

Variable	Total (N = 375)	Recurrence (N = 21)	No Recurrence (N = 354)	*P* Value
Age	61.5 ± 13.6	68.0 ± 8.4	61.1 ± 13.7	.023[a*]
Sex				.18[b]
Female	197 (52.5)	14 (7.1)	183 (92.9)	
Male	178 (47.5)	7 (3.9)	171 (96.1)	
Tumor size (mm)				
Height	3.9 ± 2.2	4.2 ± 2.1	3.9 ± 2.2	.62[a]
LBD	11.0 ± 3.3	13.0 ± 3.9	10.9 ± 3.2	.006[a*]
Distance to fovea (≥3 mm)				.25[b]
No	135 (36.0)	10 (7.4)	125 (92.6)	
Yes	240 (64.0)	11 (4.6)	229 (95.4)	
Distance to disc (≥3 mm)				.17[b]
No	95 (25.3)	8 (8.3)	87 (91.7)	
Yes	280 (74.7)	13 (4.6)	267 (95.4)	
Isotope				.20[b]
125-I	335 (89.3)	17 (5.1)	318 (94.9)	
106-Ru	40 (10.7)	4 (10.0)	36 (90.0)	
Ultrasound localization				.022[b*]
No	59 (15.7)	7 (11.9)	52 (88.1)	
Yes	316 (84.3)	14 (4.4)	302 (95.6)	
TTT				.019[b*]
No	340 (90.7)	16 (4.7)	324 (95.3)	
Yes	35 (9.3)	5 (14.3)	30 (85.7)	

LBD = largest basal diameter; TTT = transpupillary thermotherapy; 106-Ru = Ruthenium-106; 125-I = Iodine-125.

Statistics presented as mean ± SD or N (row %).

P values: [a]*t* test; [b]Pearson χ^2 test. Asterisk indicates statistical significance.

statistically significant positive predictive factors of recurrence, with a *P* value of .008 and .004, respectively (Table 3). Overall, 35 patients (47.3%) with juxtapapillary uveal melanoma, defined as a posterior tumor margin located within 1.5 mm from the optic disc, had received adjuvant TTT at the time of brachytherapy. Among the 21 cases of local recurrence, only 6 were treated by a combination of brachytherapy and TTT. Patients who had received adjuvant TTT at the time of the brachytherapy were more likely to have recurrence (P = .037) (Table 4). Conversely, juxtapapillary tumor location (1.5 mm or less from optic disc) was not found as a recurrence risk factor.

• **MANAGEMENT OF LOCAL RECURRENCE:** The majority of patients (9 patients, 75%) who experienced marginal tumor recurrence were managed conservatively with additional TTT or a repeat brachytherapy; only 3 patients underwent enucleation. Of those who had central recurrence (14.3%), 2 patients were treated with a repeat brachytherapy and 1 patient opted for enucleation because of the absence of functional vision in the affected eye. Diffuse tumor recurrences were treated by enucleation except in 1 patient who had a repeat brachytherapy owing to monocular status. All the patients with extraocular extension or ring pattern of recurrence were enucleated (Table 5).

A total of 11 patients (52.4%) with local tumor recurrence underwent enucleation at the end of the follow-up: 9 patients for whom initial brachytherapy had failed and 2 patients after failure of a salvage TTT for a marginal recurrence. The mean and median time between brachytherapy and enucleation for local treatment failure was 25 months and 17 months, respectively. A repeat brachytherapy was administered in 7 patients, with a 100% overall eye preservation rate. The median interval between the initial and repeat brachytherapy was 27 months (mean 49 months). None of these patients had developed further recurrence, painful eye, or neovascular glaucoma after a median follow-up of 45 months (range 7–108 months; mean 51 months). Of those who had local recurrence (21 patients), 10 patients (48%) developed liver metastasis at a median interval of 32.5 months (mean 40.3 months; range 18–100 months).

DISCUSSION

TO THE BEST OF OUR KNOWLEDGE, THE PRESENT STUDY IS the first to investigate and describe the chronological and anatomic patterns of local failure after brachytherapy in a large single cohort of patients followed with a standardized protocol. Our results showed that local treatment failure following brachytherapy for posterior uveal melanoma is uncommon (21 patients, 5.6%), with 5-year estimated local recurrence rate of 6.6%. This is comparable to the Ophthalmic Oncology Task Force study (7.3%), COMS data (10.3%), and other studies (Table 2).[1,13] Nonresponse to brachytherapy as indicated by progressive growth observed within 3 months following brachytherapy (2 patients, 0.5%) and early recurrence (5 patients, 1.3%) are very uncommon and has been reported previously 6 months after initial treatment in 5 of the 3217 patients (0.2%) with ciliary body and choroidal melanoma included in a multicenter shared database.[13] Conversely, Papageorgiou and associates reported that 13 of 189 patients (6.9%) with uveal melanoma showed nonresponse following ^{106}Ru brachytherapy, but the authors defined nonresponse as an increase in tumor height in the absence of an initial period of tumor control within the first year of treatment. Our 2 nonresponse cases were managed successfully by repeat brachytherapy (Figure 3).

Overall, the median time to recurrence was 18 months (range 4–156 months). The majority (90.5%) of the recurrences occurred within the first 5 years (Figure 1).

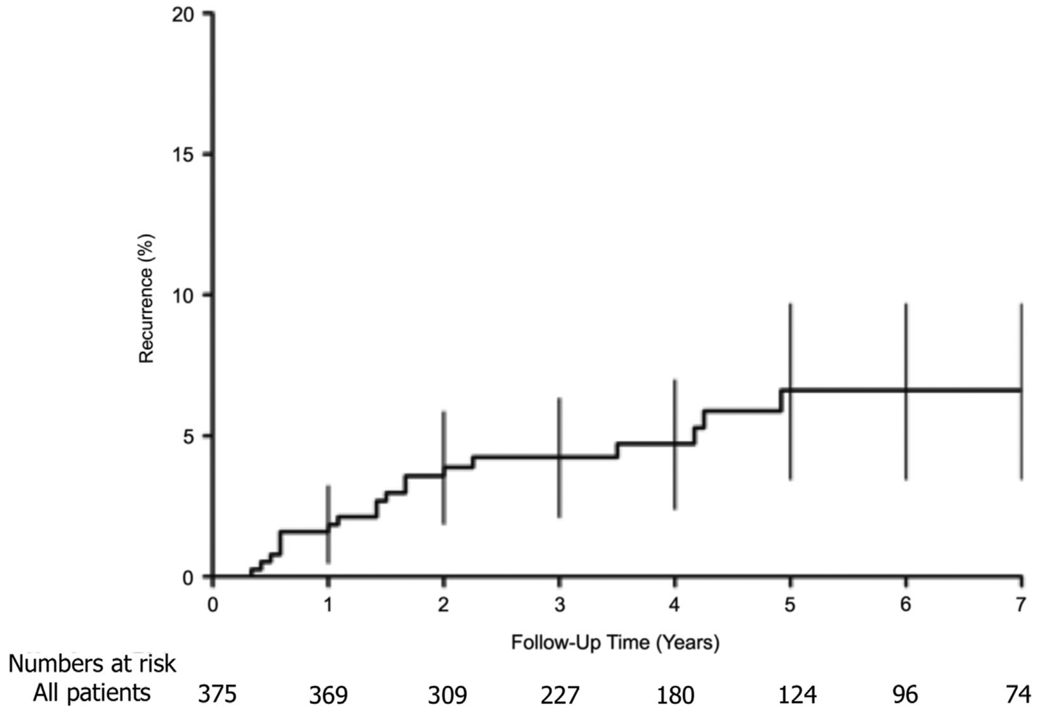

FIGURE 1. Time to local recurrence. Time to recurrence for the full cohort. Vertical lines represent 95% confidence limits for the estimate.

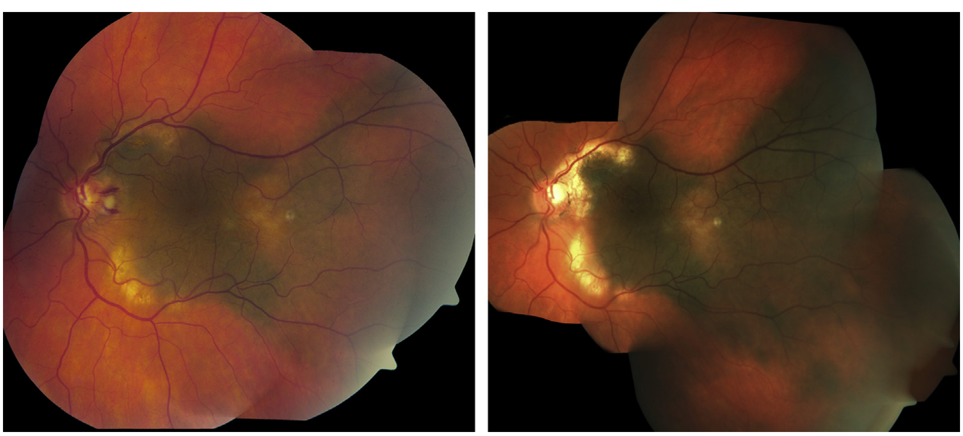

FIGURE 2. Anterior marginal recurrence. (Left) A 65-year-old man with a choroidal melanoma measuring 12 × 7 mm in basal dimensions, touching 5 clock hours of the optic disc. The patient was treated by I-125 brachytherapy and transpupillary thermotherapy. (Right) Thirteen months later, the tumor basal dimensions were 14 × 12 mm and there was evidence of anterior marginal recurrence.

Although several studies have reported very late local tumor recurrence up to 15 years after brachytherapy, late recurrence remains rare and accounts for only 2 cases in our study (more than 5 years after brachytherapy).[13,16,17,19] Owing to the low risk of recurrence beyond 5 years, less intense monitoring may be considered beyond this follow-up period.

The predominant site of recurrence was at the tumor margin (12 patients, 57.1%). Marginal recurrence of uveal melanoma has been reported as the predominant site of recurrence in other studies.[10,20] In addition to marginal recurrence, Caujolle and associates also observed 3 other types of recurrence following proton beam therapy: global, distant, and extrascleral extension.[20] In our study, recurrence may develop in a central or diffuse pattern. It is important to make the distinction between these 2 types of recurrence because diffuse recurrences including circumferential ring pattern are best treated by enucleation at the

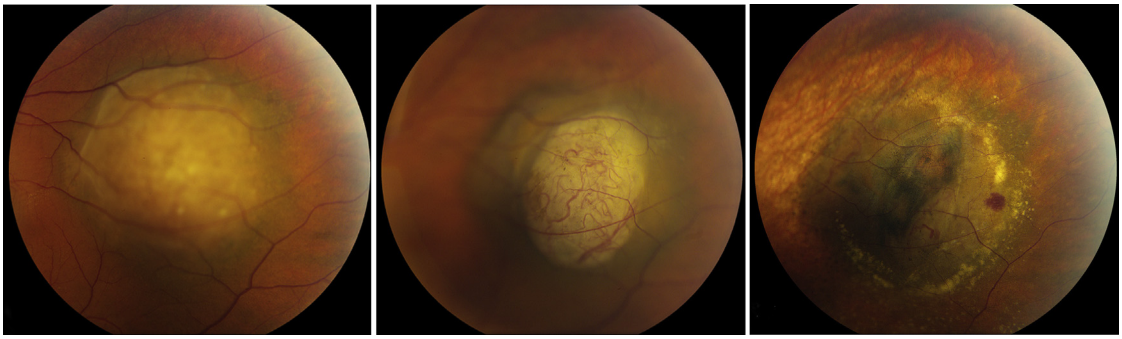

FIGURE 3. Central recurrence. (Left) A 53-year-old man with a choroidal melanoma in temporal quadrant. (Center) Three months post-brachytherapy, a central nonresponse was observed. (Right) More than 1 year after a second brachytherapy, the tumor showed regression and surrounding chorioretinal atrophy was noted.

TABLE 3. Analysis of Univariate Continuous Predictors of Local Recurrence in Patients With Posterior Uveal Melanoma

Parameter	Hazard Ratio (95% CI)	*P* Value
Age (y)	1.05 (1.01, 1.09)	.008
Apical dose	0.96 (0.89, 1.04)	.32
Distance to fovea (mm)	1.00 (0.91, 1.10)	.96
Distance to disc (mm)	0.95 (0.84, 1.06)	.35
Tumor height (mm)	1.09 (0.91, 1.31)	.34
Tumor (LBD, mm)	1.23 (1.07, 1.41)	.004

CI = confidence interval; LBD = largest basal diameter.

outset,[21,22] whereas brachytherapy appears to be an effective treatment for central recurrences according to our data. Although uncommon, extraocular recurrence is also reported as a site of recurrence.[13,23] Contrary to previous studies examining local recurrence following proton beam therapy or cobalt 60 episcleral plaque therapy, noncontiguous local recurrence was not observed in our study.[20,24]

The detailed classification of the different anatomic types of recurrence highlights the relevance of diagnostic tools, such as ultrasonography (including ultrasound biomicroscopy) to detect ring or extraocular extension. The overall patterns of local recurrence seem similar to those reported following proton beam therapy.[20] Among many plausible reasons for marginal recurrence, imprecise placement of brachytherapy implant can be minimized or avoided. Recent studies have suggested that ultrasonographic confirmation of the adequate placement of the brachytherapy implant may be associated with lower risk of tumor recurrence.[25–27] In our study, ultrasonographic confirmation had been used in 8 of the 12 patients (66.7%) who developed marginal recurrence.

Univariate analysis identified 3 statistically significant risk factors: advanced age, LBD, and the use of adjuvant TTT. Many authors have previously identified a significant correlation between greater LBD and local tumor recurrence.[1,5,10,16,18] The COMS study also found that advanced age was an independent predictor of recurrence, which is consistent with our results.[1] Juxtapapillary tumor location was not detected as a risk factor in our study, although higher 5-year tumor recurrence rate of 14% has been previously reported for juxtapapillary tumors treated with plaque and TTT.[28] As the main indication of the use of adjuvant TTT in the present study was juxtapapillary location, we consider that our number of recurrences might be too low to capture the juxtapapillary tumor location as a significant predictor of recurrence. COMS study did not identify juxtapapillary location as a risk factor for recurrence owing to the exclusion of many juxtapapillary melanomas from the COMS study.[1] A large cohort study will be needed to further clarify an association between local recurrence and the juxtapapillary location of the uveal melanoma.

Contrary to published reports of higher recurrence with the use of ^{106}Ru isotope as compared to ^{125}I isotope,[5,6,17] the type of isoptope used was not a risk factor for recurrence in our study. Several factors can explain the observed discrepancy, including our small number of cases treated with ^{106}Ru (40 patients, 10.7%), limiting their use to tumors less than 5 mm in height, and the similar apical dose as compared to ^{125}I episcleral plaque.

The treatment strategies of local recurrence are strongly dependent on the location of the recurrence. For patients with ring or extraocular recurrence, enucleation was considered as the best treatment. For patients with diffuse or central pattern of recurrence, enucleation or a repeat brachytherapy both seem to be appropriate treatment options, depending on the present visual status or future visual potential. Marginal recurrences were managed with repeat radiation, TTT, or enucleation. Of our 7 recurrent cases treated with repeat brachytherapy, 100% retained their eye at the most recent follow-up. The results were less successful in patients who received additional TTT as

TABLE 4. Analysis of Univariate Categorical Predictors of Local Recurrence in Patients With Posterior Uveal Melanoma

Group	Level	N	Recurrence at 12 Months	Recurrence at 60 Months	Hazard Ratio (95% CI)	*P* Value
Gender	Male	178	98.3 (96.5, 100.0)	95.1 (91.4, 99.0)	1.00	.25
	Female	197	98.0 (96.0, 100.0)	92.0 (87.4, 96.8)	1.70 (0.69, 4.23)	
Isotope	106-Ru	40	97.5 (92.8, 100.0)	91.4 (82.5, 100.0)	1.00	.51
	125-I	335	98.2 (96.8, 99.6)	93.4 (90.0, 96.9)	0.69 (0.22, 2.13)	
Adjuvant TTT	No	340	98.2 (96.9, 99.6)	95.1 (92.6, 97.8)	1.00	.037
	Yes	35	97.1 (91.8, 100.0)	79.2 (63.5, 98.8)	2.91 (1.06, 7.94)	
Ultrasound	No	59	96.6 (92.1, 100.0)	91.3 (84.2, 98.9)	1.00	.21
	Yes	316	98.4 (97.1, 99.8)	93.4 (89.8, 97.2)	0.54 (0.20, 1.41)	
Overall		375	98.1 (96.8, 99.5)	93.4 (90.3, 96.5)		

CI = confidence interval; TTT = transpupillary thermotherapy; 106-Ru = Ruthenium-106; 125-I = Iodine-125.

TABLE 5. Local Failure Patterns of Posterior Uveal Melanoma and Their Management (N = 21)

Patterns	Subtype	N	Mean Time (Months)	Treatment	Response
Chronological	Nonresponse	2	3	Brachytherapy	Regression 100%
	Early	5	9	TTT (1)	Regression 100%
				Enucleation (4)	
	Delayed	12	27.3	Brachytherapy (4)	Regression 100%
				TTT (3)	Regression 33%
				Observation (1)	
				Enucleation (4)	
	Late	2	106.5	Brachytherapy	Regression 100%
				Enucleation	
Anatomic	Marginal	12	39.7	Observation (1)	
				TTT (4)	Regression 50%
				Brachytherapy (4)	Regression 100%
				Enucleation (3)	
	Central	3	16	Brachytherapy (2)	Regression 100%
				Enucleation (1)	
	Diffuse	3	9	Brachytherapy (1)	Regression 100%
				Enucleation (2)	
	Ring	2	12	Enucleation	
	Extraocular	1	17	Enucleation	

TTT = transpupillary thermotherapy.

secondary treatment, eventually resulting in enucleation in 50% of cases.

In conclusion, we describe the chronological and anatomic patterns of local tumor failure among patients with posterior uveal melanoma treated by episcleral brachytherapy. Our results indicate that local recurrence is predominantly at the tumor margins and is unlikely to develop beyond 5 years after surgery. Follow-up intervals can be adjusted to reflect time to recurrence. Most of the eyes with recurrent tumor can be salvaged by conservative methods. Repeat brachytherapy for local treatment failure can result in satisfactory outcome.

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REFERENCES

1. Jampol LM, Moy CS, Murray TG, et al. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS report no. 19. *Ophthalmology* 2002;109(12):2197–2206.
2. Finger PT. Radiation therapy for choroidal melanoma. *Surv Ophthalmol* 1997;42(3):215–232.
3. Shields JA, Shields CL. Management of posterior uveal melanoma: past, present, and future: the 2014 Charles L. Schepens lecture. *Ophthalmology* 2015;122(2):414–428.
4. Chang MY, McCannel TA. Local treatment failure after globe-conserving therapy for choroidal melanoma. *Br J Ophthalmol* 2013;97(7):804–811.
5. Rouberol F, Roy P, Kodjikian L, Gerard JP, Jean-Louis B, Grange JD. Survival, anatomic, and functional long-term results in choroidal and ciliary body melanoma after ruthenium brachytherapy (15 years' experience with beta-rays). *Am J Ophthalmol* 2004;137(5):893–900.
6. Barker CA, Francis JH, Cohen GN, et al. (106)Ru plaque brachytherapy for uveal melanoma: factors associated with local tumor recurrence. *Brachytherapy* 2014;13(6):584–590.
7. Vonk DT, Kim Y, Javid C, Gordon JD, Stea B. Prescribing to tumor apex in episcleral plaque iodine-125 brachytherapy for medium-sized choroidal melanoma: a single-institutional retrospective review. *Brachytherapy* 2015;14(5):726–733.
8. Perez BA, Mettu P, Vajzovic L, et al. Uveal melanoma treated with iodine-125 episcleral plaque: an analysis of dose on disease control and visual outcomes. *Int J Radiat Oncol Biol Phys* 2014;89(1):127–136.
9. Leonard KL, Gagne NL, Mignano JE, Duker JS, Bannon EA, Rivard MJ. A 17-year retrospective study of institutional results for eye plaque brachytherapy of uveal melanoma using (125)I, (103)Pd, and (131)Cs and historical perspective. *Brachytherapy* 2011;10(4):331–339.
10. Damato B, Patel I, Campbell IR, Mayles HM, Errington RD. Local tumor control after 106Ru brachytherapy of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2005;63(2):385–391.
11. Gaspar de Souza Neves DF, Ferragut MA, Ferreira DV, et al. Reirradiation of the eye with plaque brachytherapy: a single institution experience report of eight consecutive patients submitted to retreatment after local relapse of malignant disease of the eye. *Brachytherapy* 2014;13(3):281–284.
12. Grange JD, Duquesne N, Roubeyrol F, et al. Double irradiation for macroscopic radioresistance or recurrence of melanomas of the posterior uvea: clinical, ballistic, therapeutic and prognostic aspects. Series of 19 cases among 462 patients. *J Fr Ophtalmol* 1999;22(10):1054–1063.
13. Ocular Oncology Task Force. Local recurrence significantly increases the risk of metastatic uveal melanoma. *Ophthalmology* 2016;123(1):86–91.
14. Vrabec TR, Augsburger JJ, Gamel JW, Brady LW, Hernandez C, Woodleigh R. Impact of local tumor relapse on patient survival after cobalt 60 plaque radiotherapy. *Ophthalmology* 1991;98(6):984–988.
15. Gunduz K, Shields CL, Shields JA, Cater J, Freire JE, Brady LW. Plaque radiotherapy of uveal melanoma with predominant ciliary body involvement. *Arch Ophthalmol* 1999;117(2):170–177.
16. Lommatzsch PK, Werschnik C, Schuster E. Long-term follow-up of Ru-106/Rh-106 brachytherapy for posterior uveal melanoma. *Graefes Arch Clin Exp Ophthalmol* 2000; 238(2):129–137.
17. Bergman L, Nilsson B, Lundell G, Lundell M, Seregard S. Ruthenium brachytherapy for uveal melanoma, 1979-2003: survival and functional outcomes in the Swedish population. *Ophthalmology* 2005;112(5):834–840.
18. Papageorgiou KI, Cohen VM, Bunce C, Kinsella M, Hungerford JL. Predicting local control of choroidal melanomas following (1)(0)(6)Ru plaque brachytherapy. *Br J Ophthalmol* 2011;95(2):166–170.
19. Char DH, Kroll S, Phillips TL, Quivey JM. Late radiation failures after iodine 125 brachytherapy for uveal melanoma compared with charged-particle (proton or helium ion) therapy. *Ophthalmology* 2002;109(10): 1850–1854.
20. Caujolle JP, Paoli V, Chamorey E, et al. Local recurrence after uveal melanoma proton beam therapy: recurrence types and prognostic consequences. *Int J Radiat Oncol Biol Phys* 2013;85(5):1218–1224.
21. Demirci H, Shields CL, Shields JA, Honavar SG, Eagle RC Jr. Ring melanoma of the ciliary body: report on twenty-three patients. *Retina* 2002;22(6):698–706. quiz 852–853.
22. Aziz HA, Modi YS, Plesec TP, Singh AD. Amelanotic iridociliary ring melanoma: a clinicopathological correlation. *Ocul Oncol Pathol* 2016;2(3):153–155.
23. Lantigua-Dorville Y, Saornil MA, Garcia-Alvarez C, Garcia-Lagarto E. Choroidal melanoma recurrence after episcleral brachytherapy and transpupillary thermotherapy. *Arq Bras Oftalmol* 2016;79(1):48–49.
24. Duker JS, Augsburger JJ, Shields JA. Noncontiguous local recurrence of posterior uveal melanoma after cobalt 60 episcleral plaque therapy. *Arch Ophthalmol* 1989;107(7): 1019–1022.
25. Tabandeh H, Chaudhry NA, Murray TG, et al. Intraoperative echographic localization of iodine-125 episcleral plaque for brachytherapy of choroidal melanoma. *Am J Ophthalmol* 2000;129(2):199–204.
26. Aziz HA, Al Zahrani YA, Bena J, et al. Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation. *Br J Ophthalmol* 2016; http://dx.doi.org/10.1136/bjophthalmol-2016-309153.
27. Chang MY, Kamrava M, Demanes DJ, et al. Intraoperative ultrasonography-guided positioning of iodine 125 plaque brachytherapy in the treatment of choroidal melanoma. *Ophthalmology* 2012;119(5):1073–1077.
28. Sagoo MS, Shields CL, Mashayekhi A, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma overhanging the optic disc in 141 consecutive patients. *Arch Ophthalmol* 2008;126(11):1515–1522.